Comparing GeoToken, G3, and GeoCLIP on Street-View Images of Serbia

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Abstract

Worldwide image geolocalization models are trained on globally distributed corpora of millions of images, but it is unclear how their design choices transfer to a single small country, where the geographic search space is narrow and training data is scarce. We adapt three recent geolocalization paradigms: GeoToken's hierarchical autoregressive token prediction, G3's tri-modal contrastive alignment with geo-verification, and GeoCLIP's contrastive backbone repurposed for direct coordinate regression. We study how they perform on a 20,366-image subset of OpenStreetView-5M restricted to Serbia, and evaluate all three through a single shared pipeline on an identical sequence-grouped split. Off-the-shelf pretrained models transfer poorly: both zero-shot GeoCLIP and zero-shot G3 perform no better than predicting the country centroid (∼100 km median error), and domain adaptation improves this five- to eightfold. On the held-out test split GeoToken performs best at 11.2 km median error, ahead of G3 (12.9 km with test-time augmentation) and GeoCLIP (20.5 km); the single largest contributor is geo-alignment encoder pretraining, worth a 32% reduction on its own. Beyond the ranking, the models divide by output type *rather than by training objective: discrete predictors are more than an order of magnitude more accurate at street level (9% vs. 0.4% within 1 km) yet fail beyond 200 km three to six times as often. Test-time augmentation, by contrast, does not follow that division: it helps in proportion to how far a prediction rests on a single embedding, which is why it is worth 6.8 km to G3 but only 0.5 km to GeoToken.*

1. Introduction

Image geolocalization, known as predicting where a photograph was taken from its pixels alone, has advanced rapidly through models trained on millions of globally distributed images [1, 12, 9]. These models are designed for a planet-scale search space, and much of their machinery reflects that: coarse-to-fine hierarchies that first separate continents, retrieval galleries spread thinly across the globe, and representations tuned to distinguish Tokyo from Paris.

It is not obvious that these choices remain appropriate when the search space collapses to a single small country. Serbia spans roughly 4° of latitude; distinguishing two towns 30 km apart is a different problem from distinguishing two continents, and the available training data is smaller by three orders of magnitude. This paper asks which geolocalization paradigms survive that change of regime.

We chose to adapt three recent models to a Serbia-only subset of OpenStreetView-5M [1] and compare them under a single evaluation protocol:

- **GeoToken** [4]: autoregressive prediction of a hierarchical S2-cell token sequence, conditioned on retrieved context.
- **G3** [9]: contrastive alignment of image, GPS, and text embeddings, followed by similarity-based geo-verification over a candidate gallery.
- **GeoCLIP** [12]: a CLIP-based contrastive backbone, which we adapt by replacing its retrieval head with direct coordinate regression. Its contrastive image–GPS formulation is not lost from the comparison: our G3 reuses GeoCLIP's location encoder, so the text-free G3 ablation of Section 5.5 is that formulation fine-tuned on Serbia.

These are successive generations of one line of work rather than three independent systems: G3 builds its location encoder on GeoCLIP's, and GeoToken takes its geo-alignment stage directly from G3, so comparing them measures what each addition is worth once the search space shrinks to a single country.

Our contributions are:

- A controlled three-way comparison of geolocalization paradigms on a single small-country dataset, with matched training budgets and one shared evaluation pipeline.
- Evidence that off-the-shelf worldwide models transfer essentially nothing to intra-country localization: zero-shot GeoCLIP and G3 both perform at centroid level, and adaptation yields a five- to eightfold improvement.

- An ablation isolating geo-alignment encoder pretraining, the single largest contributor to GeoToken's performance (32% median reduction), and a selection-strategy comparison in which G3's geo-verification, reused as a reranker for GeoToken's candidate pool, outperforms log-probability selection.

- A quantified precision/robustness trade-off that follows from *output type* (discrete versus continuous) rather than from training objective, and is invisible in median error alone.

- A demonstration that conclusions drawn at short training budgets can reverse once budgets are matched, with two concrete instances.

2. Related Work

Classification and retrieval. Early approaches framed geolocalization as retrieval against a geotagged database [7], or as classification over a discretized grid of cells [13]. Both are limited by their spatial resolution: classification cannot predict locations far from a cell centre, and retrieval cannot predict a location absent from its gallery. We quantify exactly this limitation in Section 5.

Contrastive alignment. GeoCLIP [12] aligns CLIP [11] image embeddings with GPS coordinates encoded through random Fourier features, trained contrastively with SimCLR-style augmentation [2]. Inference ranks a query image against a fixed gallery of GPS candidates. StreetCLIP [5] takes a zero-shot variant of this idea using text-encoded location prompts, and PIGEON [6] combines CLIP features with semantic geocells.

Hybrid and generative approaches. G3 [9] extends contrastive alignment to three modalities (image, GPS, and reverse-geocoded text) and adds a multimodal LLM stage that generates candidate coordinates, followed by similarity-based verification. GeoToken [4] instead frames the task as sequence generation over hierarchical S2 cells, decoding coarse-to-fine. PLONK [3] takes a generative approach via diffusion on the Earth's surface. We exclude PLONK as a baseline because it is trained on OpenStreetView-5M, which contains our evaluation images.

Parameter-efficient adaptation. LoRA [8] injects low-rank adapters beside frozen weights, enabling fine-tuning of large backbones at a fraction of the parameter cost. We use it to compare parameter-efficient adaptation against full fine-tuning of the last transformer blocks, a comparison complicated at small data scale by catastrophic forgetting [10], where high-magnitude updates overwrite the features acquired during pretraining.

3. Dataset

Source and scope. We use OpenStreetView-5M [1], a corpus of street-level photographs contributed to OpenStreetMap, filtered to images captured in Serbia. This yields **20,366 images** from **81 contributors**, covering everyday roads, intersections, and rural areas rather than landmarks. Serbia offers a compact but varied target: Pannonian plains in the north, mountainous terrain in the south, and a road network dense enough to support retrieval-based methods.

Sequence-grouped splitting. OSV5M images arrive in *sequences*: bursts of frames captured seconds apart as a contributor drives a route. Our data contains 3,918 sequences averaging 5.2 images each, so consecutive frames within a sequence are near-duplicates. A naive per-image random split scatters these across partitions; we measured that 86–93% of validation and test images then had a near-identical twin in training, allowing a model to score well by recognizing a specific scene rather than by learning geography. Because this would inflate retrieval-based methods specifically, it would bias the very comparison this paper makes.

We therefore split by whole sequence, so every frame of a burst lands in exactly one partition: **16,294 train / 2,046 validation / 2,026 test**, with zero sequence overlap. Sequences are shuffled with a fixed `seed=42` and the resulting partition is stored as index files, so all three models train and evaluate on an identical split.

Residual correlation. Sequence grouping removes near-duplicate frames but not all dependence between partitions. For each validation image we located its geographically nearest training image and found that none share a sequence (0/2,046), but 61.2% share a *contributor* and 46.0% were captured within one day. With only 81 contributors, identical camera, weather, and lighting conditions are therefore not fully held out. A contributor-grouped or geographically blocked split would be stricter. The gap between the retrieval ceiling and achieved performance (Section 5) suggests this residual correlation is not substantially exploited.

Preprocessing. Images are resized to 224×224 for evaluation. Training applies the augmentation policy used by **GeoCLIP's** own pretraining [2]: random crop ($256 \rightarrow 224$), horizontal flip (p=0.5), colour jitter, rotation ($\pm 15°$), and random grayscale (p=0.05), with CLIP normalization statistics. For **GeoToken**, coordinates are converted to 21-token hierarchical S2 sequences; for **G3**, place descriptions are constructed from OSV5M's reverse-geocoded metadata fields, yielding 298 unique strings over the training split.

4. Method

All three models share a frozen or adapter-modified CLIP ViT-L/14 [11] visual backbone. What differs is the loss function (training objective) and the prediction mechanism, summarized in Table 1.

Model	Loss	Prediction
GeoToken	token cross-entropy	autoregressive S2 decode
G3	InfoNCE (3 modalities)	argmax similarity over gallery
GeoCLIP	Haversine + L1	direct (ϕ, λ) regression

Table 1. The three paradigms compared. All share a CLIP ViT-L/14 backbone.

4.1. GeoToken: hierarchical token generation

A coordinate is encoded as a sequence of $L{=}21$ S2 tokens: a face token $t_0 \in \{0, \ldots, 5\}$ and 20 quadrant tokens $t_i \in \{0, 1, 2, 3\}$, each refining the previous level. A shared prefix of length ℓ means two locations fall in the same S2 cell at level $\ell{-}1$, so token-space distance corresponds to real-world proximity.

Given a query image, we retrieve its $M{=}15$ nearest neighbours from a gallery of training-image embeddings by cosine similarity. An encoder Transformer processes the query embedding together with the retrieved neighbours' embeddings and token sequences; a causal decoder then emits the query's token sequence one position at a time, modelling $P(t_s \mid t_{<s}, \mathrm{Enc}(\mathbf{X}))$.

Training uses cross-entropy per position with the position-dependent weighting of [4], which places greater weight on coarse levels where errors are geographically costlier.

Encoder pretraining. The paper's encoder block emits $\mathrm{concat}(e^{\mathrm{raw}}, e^{\mathrm{text}}, e^{\mathrm{gps}})$, where the latter two come from the geo-alignment pretraining of their Section III-B. We supply this using our own trained G3 image tower, concatenating raw CLIP features (768-d) with G3-aligned features (512-d), each L2-normalised, into a 1280-d embedding. This is a two-view approximation of their three-view concatenation, since our G3 uses one shared image projection rather than separate text- and GPS-aligned ones. Section 5 reports its effect.

Remaining deviations. We compress each neighbour's 21 token embeddings into a single pooled vector fused with its image embedding, giving an encoder sequence of $1{+}M$ rather than $1{+}M(1{+}L)$, or 16 positions instead of 331. On a single consumer GPU we judged the geo-alignment encoder above the better use of that budget, and Section 5 bears this out. We also do not evaluate beam search or the MLLM-based selection strategy.

4.2. G3: tri-modal alignment and verification

G3 comprises three stages: geo-alignment, geo-diversification, and geo-verification. We implement the first and third.

Geo-alignment trains image, GPS, and text encoders into a shared space with a symmetric InfoNCE objective,

$$\mathcal{L} = \tfrac{1}{2}\left(\mathcal{L}_{\mathrm{img,txt}} + \mathcal{L}_{\mathrm{img,gps}} + \mathcal{L}_{\mathrm{txt,img}} + \mathcal{L}_{\mathrm{gps,img}}\right), \quad (1)$$

where GPS coordinates pass through a projection followed by multi-scale random Fourier features, and text descriptions are obtained by reverse geocoding the coordinate (e.g. *"A photo taken from Bela Palanka, Pirotski Okrug, Central Serbia"*). Following [12], a queue of 4,096 past GPS coordinates supplies additional negatives, since a batch of 32 alone provides too few. Because several images in a batch can share a description, duplicate descriptions are masked out of the text denominator; treating them as negatives would push apart images that belong together.

Geo-verification predicts by selecting the gallery coordinate whose GPS embedding is most similar to the query image embedding.

Deviations. We omit geo-diversification, which uses a multimodal LLM to generate candidate coordinates. Our justification is empirical: the candidate pool already contains a near-perfect answer for most queries (Section 5), so the limiting factor is candidate *selection*, not generation. We also use a flatter text hierarchy than the original, since country and continent are constant within Serbia.

4.3. GeoCLIP: retrieval replaced by regression

GeoCLIP natively performs contrastive image-GPS alignment followed by retrieval over a fixed gallery of $\sim$100K worldwide GPS candidates. We replace that retrieval head with direct coordinate regression.

Its contrastive form is still measured. Fine-tuned on Serbia and retrieving over Serbian coordinates, GeoCLIP's own formulation is exactly our G3 model with the text branch removed. We do not reimplement it here; it is reported as the text-free arm of Section 5.5.

Regression adaptation. Motivated by the gallery-sparsity measurement reported in Section 5, we discard the location encoder and gallery entirely and attach a regression head ($512{\to}1024{\to}512{\to}256{\to}2$, BatchNorm/GELU/dropout, final sigmoid) mapping into Serbia's bounding box by construction. Training minimizes the loss function: $\mathcal{L} = 0.05\,\mathcal{L}_{\mathrm{L1}} + 0.95\,\mathcal{L}_{\mathrm{Hav}}$. The motivation is gallery sparsity: only **70** of GeoCLIP's 100,000 gallery points fall inside Serbia – a density that caps any retrieval method built on that gallery, as we quantify in Section 5. We compare full fine-tuning of the last two ViT blocks against LoRA adapters [8] of varying rank and scope.

4.4. Evaluation protocol

All models, baselines and adapted alike, are scored by one shared pipeline computing Haversine error, median, mean, and accuracy at 1, 25, 200, 750, and 2500 km. Model selection uses validation median error; the test partition is evaluated once, after all decisions are frozen.

Reference	Median (km)	Accuracy (%) 1km	25km	200km
Centroid (image ignored)	99.7	0.00	1.96	96.24
GeoCLIP zero-shot, global	439.6	0.05	0.24	14.57
GeoCLIP zero-shot, Serbia	104.5	0.64	5.43	87.78
G3 pretrained, zero-shot	100.6	0.00	5.47	84.90
Oracle: GeoCLIP gallery	*20.45*	*3.96*	*60.31*	*100.0*
Oracle: train gallery	*0.35*	*72.24*	*100.0*	*100.0*

Table 2. Reference points on validation (2,046 images). Oracles select the closest gallery point to ground truth and bound any retrieval method using that gallery; they are not predictors.

Ver.	Adaptation	Train.	Ep.	Median	25km
v1	full unfreeze	28.1M	35	**17.7**	**61.49%**
v2	LoRA $\{q, v\}$ r=8	3.7M	35	24.0	51.61%
v3	LoRA $\{q, k, v, o\}$ r=16	6.1M	35	24.2	51.42%
v4	+ warmup, lr 5e-5	6.1M	35	22.7	54.15%

Table 3. GeoCLIP regression adaptation on validation, every configuration trained for the same 35 epochs. Median in km.

5. Experiments

5.1. Reference points and ceilings

Table 2 establishes what any model must beat and what retrieval can achieve at best.

Three observations follow. First, **every off-the-shelf model performs at centroid level.** Once given the same "it is in Serbia" prior our models have by construction, zero-shot GeoCLIP (104.5 km) and zero-shot G3 (100.6 km) have similar performance to ignoring the image entirely (99.7 km). Both retain weak signal (roughly 5.4% accuracy at 25 km against the centroid's 1.96%) but they scatter predictions widely enough that it does not survive into the median, and both are in fact *worse* than the centroid at 200 km.

Second, **gallery density, not the retrieval paradigm, is the binding constraint.** A perfect retriever restricted to GeoCLIP's shipped gallery could not beat 20.45 km, because only 70 of its points lie inside Serbia. Retrieving over the 16,294 training coordinates instead has a ceiling of 0.35 km. This justifies both of our seemingly opposite design choices: abandoning retrieval for GeoCLIP's sparse gallery, and retaining it for a dense in-domain one.

Third, that 0.35 km ceiling explains why we omit G3's MLLM stage: a near-optimal candidate is already present for 72% of queries, so generating more candidates addresses a problem we do not have.

5.2. Overview of ablations

We report three ablation studies, each isolating one design decision while holding the split, evaluation pipeline, and training budget fixed: **(I)** GeoCLIP's adaptation strategy (adapter rank, scope, and schedule against full fine-tuning); **(II)** GeoToken's components (geo-alignment encoder, selection strategy); and **(III)** G3's text modality. Test-time augmentation is reported on and off in Table 5.

5.3. Ablation I: GeoCLIP adaptation strategy

Table 3 traces a progression from full fine-tuning of the last two transformer blocks to increasingly constrained LoRA configurations, isolating the effect of adapter rank, adapter scope, and learning-rate schedule.

Two findings run against expectation. First, **we observed no catastrophic forgetting.** Fully unfreezing the last two ViT blocks at 10^{-4} on only 16,294 images is the regime in which pretrained features are most at risk of being overwritten [10]; in our setting it was instead the *best* configuration. Second, **LoRA did not match full fine-tuning** at any rank or scope we tested, though it was consistently more resistant to overfitting meaning that training and validation loss plateaued together, whereas full fine-tuning showed the usual divergence.

Within LoRA, **rank and scope made no measurable difference.** At a matched 35 epochs the smallest adapter (v2, r=8 on $\{q, v\}$, 3.7M parameters) reaches 24.0 km on validation against the largest (v3, r=16 on $\{q, k, v, o\}$, 6.1M) at 24.2 km. The only change that moved LoRA was the learning rate and warmup schedule (v4, 22.7 km), and extending v2 from 20 to 35 epochs was worth just 0.5 km.

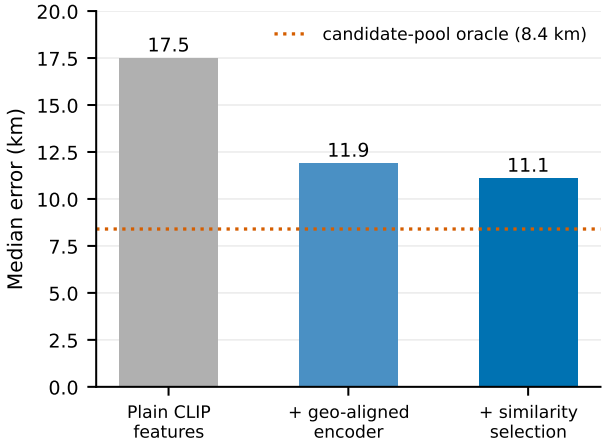


Figure 1. GeoToken components, validation median error, added cumulatively left to right. The dashed line is the ideal-selector ceiling: the error if an oracle picked the best candidate from the pool the decoder already produces. Most of the gain comes from the retrieval context; the remaining gap to the ceiling is a candidate-selection problem, not a generation one.

GeoToken variant	Median (km)	1km	25km
Plain CLIP features	17.5	6.30	57.09
+ geo-aligned encoder	**11.9**	**8.31**	**64.86**
Selection strategy, K=10, T=0.7			
Greedy	11.9	8.31	64.86
Log-probability	11.6	8.55	65.15
Similarity-based (G3)	**11.1**	**10.02**	**66.13**
Ideal Selector (oracle)	*8.4*	*16.42*	*70.58*

Table 4. GeoToken components, validation. Median in km, accuracies in %.

5.4. Ablation II: GeoToken components

Table 4 isolates the two GeoToken components we added, and Figure 1 shows the same progression against the candidate-pool ceiling.

Geo-alignment is the single largest contributor, reducing median error by 32% (17.5→11.9 km). This directly corroborates the original authors' emphasis on encoder pretraining, and shows the effect persists at single-country scale where the encoder need not separate continents.

Selection strategy matters more than decoding. Sampling ten candidates produces a pool whose best member averages 8.4 km, with twice the street-level accuracy of greedy decoding, but log-probability selection recovers only 9% of that headroom. Similarity-based selection, implemented by applying our G3 geo-verification stage to GeoToken's candidate pool, recovers 23%. The difference is informative: log-probability measures the decoder's confidence in its own token sequence, which carries no information about whether the resulting *location* suits the image, whereas similarity scoring supplies independent evidence. The remaining 2.7 km gap is what the paper's reward-model and MLLM-judge strategies target.

5.5. Ablation III: G3's text modality

Removing the text branch reduces G3 to image–GPS alignment, i.e. GeoCLIP's formulation. At a matched 35 epochs the text-free arm reaches **17.7 km** median against **18.1 km** with text, and the two are indistinguishable at every threshold: 8.31 vs. 8.16% at 1 km, 56.01 vs. 56.11% at 25 km, and 95.85 vs. 95.89% at 200 km. The differences are fractions of a point and do not share a direction, so we read the text modality as contributing nothing measurable in this setting. The same holds at 20 epochs (19.25 vs. 19.41 km), so this is not an artifact of budget.

This follows from what the descriptions can encode here. Reverse geocoding our training split yields only 298 unique strings across 16,294 images, so roughly 55 images share each description. Place names separate cities; inside a single small country nearly every pair of images already shares one. G3's third view is close to constant on this data, which is regime in which a contrastive term can carry no signal.

Model	Median (km)	Mean (km)	Accuracy (%) 1km	25km	200km
Centroid	*84.9*	*94.4*	*0.00*	*4.00*	*94.37*
GeoCLIP zero-shot (RS)	*96.8*	–	*0.54*	*6.76*	*89.24*
Oracle: GeoCLIP gallery	*22.3*	–	*4.10*	*57.90*	*100.0*
GeoToken + TTA	**11.2**	35.1	**9.67**	64.41	97.14
GeoToken (geo-aligned)	11.7	35.7	8.98	**64.71**	96.94
G3 + TTA	12.9	42.5	9.33	60.02	95.21
G3	19.7	50.9	7.26	53.16	95.01
GeoCLIP v1 + TTA	20.5	**30.2**	0.79	57.50	**99.46**
GeoCLIP v1	22.3	33.6	0.39	54.69	99.11

Table 5. Final comparison on the held-out **test** split (2,026 images), evaluated once after all model and hyperparameter decisions were frozen. Italic rows are reference points, not predictors. RS = Serbia-restricted gallery.

5.6. Effect of training budget

Two separate comparisons reversed or materially changed once budgets were matched. At 20 epochs v3 outperformed v4 (24.7 vs. 27.2 km); at 35 epochs the ranking inverted (24.2 vs. 22.7 km). Configuration v1 appeared to plateau at 20.4 km after 15 epochs, but reached 17.7 km under an identical configuration given a longer schedule.

The v1 case is instructive: its apparent 20.4 km ceiling was an artifact of a cosine schedule annealed to near-zero by epoch 11, not a capacity limit. We therefore report the training budget alongside every ablation result.

5.7. Main comparison

GeoToken is the strongest model at 11.2 km with test-time augmentation and 11.7 km without, ahead of G3 with augmentation (12.9 km) and GeoCLIP (20.5 km). It is also the only model whose test performance matches its validation performance (11.9→11.7 km). GeoCLIP v1 degraded by 4.6 km (17.7→22.3) and G3 by 1.6 km (18.1→19.7). We attribute GeoCLIP's gap to checkpoint selection: its best epoch was chosen from a validation curve whose minimum did not transfer, whereas GeoToken's curve was comparatively stable.

Note also that GeoCLIP lands at exactly the retrieval oracle's 22.3 km. The regression model therefore performs precisely as well as a *perfect* retriever restricted to GeoCLIP's shipped gallery – further evidence that discarding that gallery, rather than improving retrieval over it, was the right decision.

The split is not retrieval versus regression. It is discrete versus continuous output. GeoToken and G3, which emit discrete predictions (S2 cells and gallery entries respectively), place 8.98% and 7.26% of predictions within 1 km against GeoCLIP's 0.39% – more than an order of

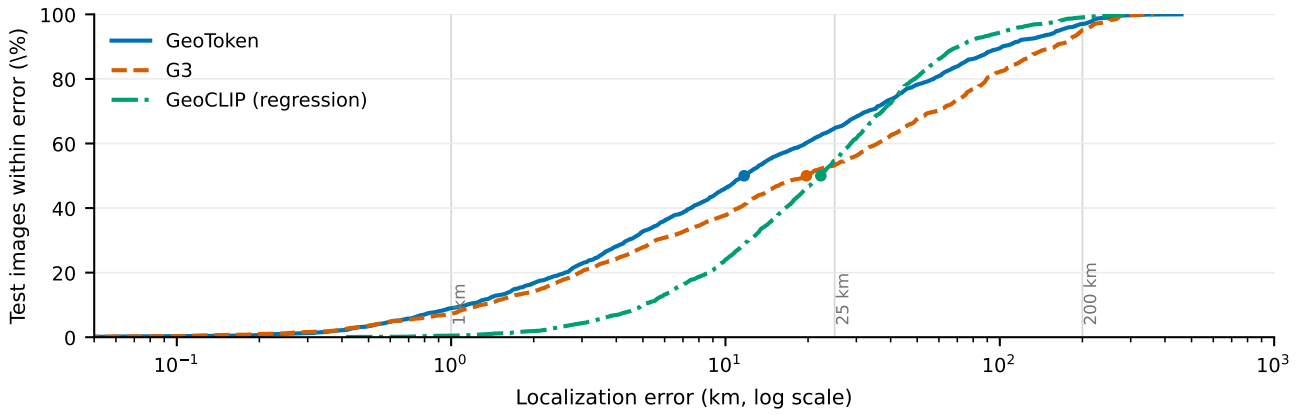


Figure 2. Cumulative error distribution on the test set: the fraction of images localized within a given error, with a marker on each curve at that model's median. All three models are shown without test-time augmentation, so the comparison is like for like; the TTA variants appear separately in Table 5. The curves cross, the discrete predictors leading at small errors and the regression head overtaking them in the tail, which is why we report accuracy at several thresholds rather than a single statistic.

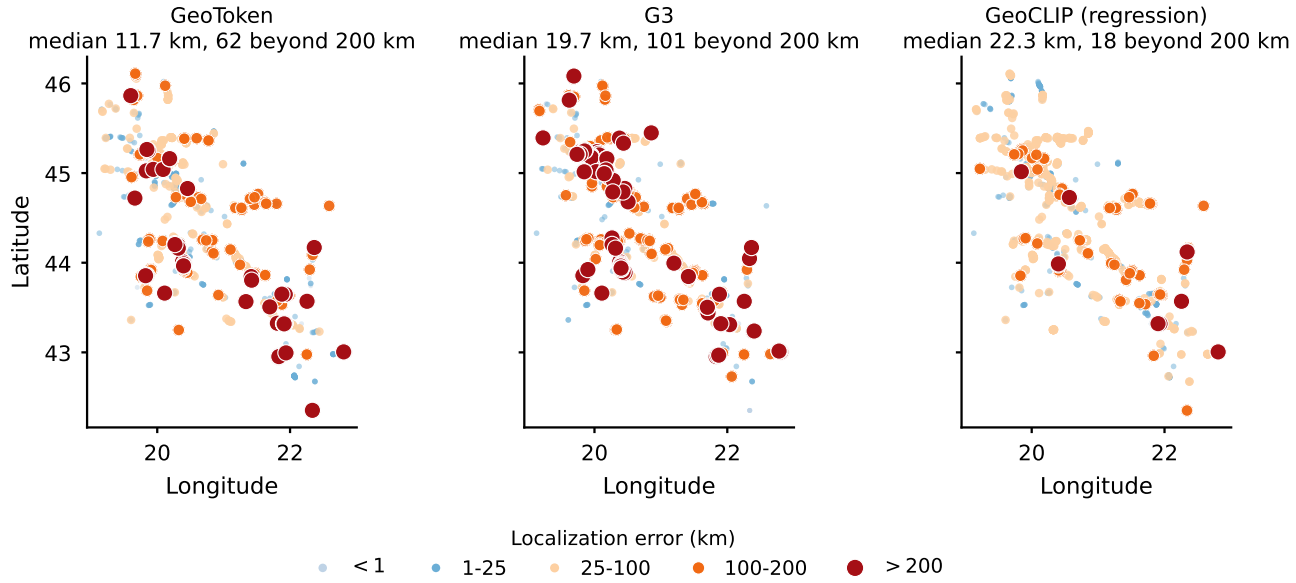


Figure 3. Test locations coloured by localization error, with failures beyond 200 km drawn large and in red so the tail is visible against the bulk; panel titles give each model's median and its number of such failures. The discrete predictors fail catastrophically several times as often as the regression head, which is the same cost that appears in Figure 2 as the crossing of the cumulative curves.

magnitude better at street level. They pay for it in the tail, failing beyond 200 km three to six times as often (3.1% and 5.0% vs. 0.9%), and their means are higher (35.7 and 50.9 km vs. 33.6 km) despite far better medians. These are the no-augmentation rows throughout, matching Figure 2.

The mechanism is structural. A discrete predictor either lands on the correct cell or selects a visually similar one hundreds of kilometres away (Figure 3), whereas a continuous head interpolates: its output moves smoothly with the image embedding, so it is rarely exact but also rarely far wrong.

We tested the obvious alternative explanation, that the sigmoid-bounded head simply retreats to safe ground near the country's centre, and rejected it. Taking each model's predicted-coordinate standard deviation as a fraction of the ground truth's, GeoCLIP measures $1.01\times$, GeoToken $1.04\times$ and G3 $1.09\times$. The regression head covers Serbia's extent as fully as the discrete models do, so its small tail follows from the smoothness of its output rather than from any contraction toward the interior. Notably GeoToken ex-

hibits this profile despite being a *generative* model rather than a retrieval one: conditioning on retrieved neighbours and emitting discrete cells is apparently sufficient to inherit retrieval's characteristic behaviour.

Test-time augmentation helps in proportion to how much a prediction rests on a single embedding. Averaging six deterministic views before prediction gains G3 **6.8 km** (19.7→12.9), GeoCLIP 1.8 km (22.3→20.5), and GeoToken only 0.5 km (11.7→11.2). The ordering is not discrete versus continuous: GeoToken is a discrete predictor and gains the least of the three, less even than the regression model.

What separates them is how concentrated each decision is. G3 predicts by a single $\arg\max$ over the gallery, so a small embedding perturbation flips the output to an entirely different coordinate. GeoCLIP's continuous head moves smoothly with its input, so averaging recovers proportionally less. GeoToken conditions on 15 retrieved neighbours and decodes a token sequence, which is already an ensemble over retrieved context, leaving little view noise for test-time augmentation to remove.

GeoToken's small gain is also confined to the fine end, improving 1 km accuracy from 8.98 to 9.67% while 25 km accuracy falls slightly, 64.71 to 64.41%. It is the only model that augmentation makes marginally worse at the coarse threshold.

6. Conclusion

We adapted three geolocalization paradigms to a single small country and compared them under matched budgets and a shared evaluation pipeline. Pretrained worldwide models transfer essentially nothing to intra-country localization, performing at centroid level until adapted. After adaptation GeoToken performs best (11.2 km test median), ahead of G3 (12.9 km) and GeoCLIP (20.5 km), with geo-alignment encoder pretraining the single largest contributor. The models divide by output type rather than training objective: discrete predictors achieve an order of magnitude better street-level precision, at the cost of six times as many catastrophic failures. Test-time augmentation cuts across that division rather than following it, helping each model in proportion to how much of its prediction rests on a single embedding.

Future work. Our GeoToken implementation still compresses each neighbour's retrieved token sequence into a single pooled vector rather than the paper's full $1{+}M(1{+}L)$ encoder input, and we did not evaluate beam search or the reward-model and MLLM-judge selection strategies: the Ideal-Selector oracle indicates 2.7 km of headroom these could target. A contributor-grouped split would eliminate the residual correlation documented in Section 3, and G3's geo-diversification stage remains untested in this regime.

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